



High energy and water usage are among causes of concern (Credit: IDTechEx)

electronics industry, facilitating the transition to greener manufacturing practices.

## Steps to sustainability

Silicon's journey in the tech industry begins with the production of large silicon ingots, sliced into thin wafers. These wafers serve as the foundation for multiple complex processes at fabrication plants. The wafer's surface is initially oxidised, followed by photolithography, which patterns the silicon surface. The wafers are then etched and metalised to form circuits and ensure conductivity. Doping modifies their electrical properties, integrating transistors and resistors before semiconductor packaging.

Simultaneously, PCB production starts with an FR4 substrate topped with copper, which is etched to create circuit patterns. Additional components, such as resistors and transistors, are attached, with PCBs potentially comprising up to 32 layers. These are meticulously assembled before final device integration.

Sustainability is now a critical focus, aiming to reduce material and energy consumption, lower costs, and minimise air pollution and electronic waste. A stark example of environmental impact is the vast 'waste island' in the ocean—comparable in size to Germany—highlighting the urgency of e-waste reduction.

Industry-wide collaboration is essential for improving

sustainability. The complexity of the manufacturing chain means that even minor adjustments can have widespread effects. Organisations such as Semiconductor Equipment and Materials International (SEMI) play a key role in fostering cooperation. A notable example is electronics company Mayerhofer's use

of secondary tin, which substitutes recycled tin of comparable quality for mined tin, demonstrating that sustainability does not necessitate quality compromise.

Achieving sustainability in semiconductor and PCB manufacturing requires a collective commitment to recycled materials and greater efficiency—paving the way for a more environmentally responsible industry.

## Semiconductor industry is making every drop count

Due to the industry's intensive water demands, efficient water management is crucial in semiconductor manufacturing. Increasing water reuse is a key strategy: reprocessed water serves multiple functions, from rinsing microprocessors with ultra-pure water to scrubbing towers and cooling systems. These practices reduce reliance on municipal supplies and improve facility water cycle efficiency.

Another critical approach is enhancing process efficiency. TSMC, for instance, uses about 170 litres of water per silicon wafer, largely due to extensive rinsing—often exceeding 30 steps per wafer. Strategies such as shortening rinse times—akin to GlobalFoundries' reduction from 10 minutes to five—along with batch wafer tools for simultaneous processing significantly curtail water and chemical consumption.

Diversifying water sources is equally important. Beyond

municipal water, some facilities collect rainwater, which—although a small fraction of total usage—contributes millions of litres annually. Coastal facilities are increasingly adopting desalination, despite its energy intensity.

Categorising wastewater efficiently enhances treatment and reuse. Semiconductor production involves diverse contaminants at different stages. For instance, United Microelectronics Corporation (UMC) classifies wastewater into up to 27 types, allowing tailored treatment. Some treated water can be repurposed for less sensitive processes, maximising resource use.

The need for ultra-pure water underscores the industry's complex water management challenges. Producing 1000 litres of ultra-pure water requires around 1700 litres of tap water, undergoing multiple processes, including UV treatment, reverse osmosis, and ultrafiltration with nano-sized pores. Balancing high purity standards with sustainability goals makes wastewater reuse difficult and essential.

## Semiconductor giants' contribution to sustainability

Efficient water and energy management is paramount in semiconductor manufacturing, mainly as many fabrication plants operate in water-scarce regions such as Taiwan and Arizona. Innovations in water recycling are proving effective. For example, NXP repurposes wastewater for cooling towers, while GlobalFoundries has halved rinse times, saving 10 million litres of water annually. Intel approaches near-complete water reuse across its operations, and Samsung has achieved a 70% recycling rate while also working to replenish extracted water.

Chemical reuse has also advanced significantly. Screen, a leading manufacturer of wet chemical tools